

Python Final Exam

Instructions:

1. This is a closed book exam.
2. Answer all questions.
3. Write readable, well-documented code.

1. (5 marks) Function to Find Common Elements

Write a Python function `find_common_elements(list1, list2)` to find all elements that appear in both `list1` and `list2`. The function returns a list of the common elements, without duplicates, sorted in ascending order. If no common elements are found, return an empty list.

Example:

```
find_common_elements([1, 2, 3, 4], [3, 4, 5, 6]) -> [3, 4]
```

```
find_common_elements([7, 8], [9, 10]) -> []
```

2. (6 marks) Function to Sort Employees by Salary

You are given a dictionary `employee_salaries` where each key is an employee's name, and each value is their salary. Define a function `rank_employees(employee_salaries)` that returns a list of employee names ranked from highest to lowest salary. If two employees have the same salary, their order does not matter.

Example:

```
rank_employees({'Alice': 50000, 'Bob': 45000, 'Eve': 50000}) -> ['Alice', 'Eve', 'Bob']
```

3. (8 marks) Recursive Function to Calculate Power

Define a recursive function `power(base, exponent)` that calculates the value of `base` raised to the

power of `exponent`. Assume that both `base` and `exponent` are non-negative integers.

Example:

`power(2, 3)` -> 8

`power(5, 0)` -> 1

4. (10 marks) Class Implementation: Library System

Define a class `LibraryBook` that represents a book in a library. Each book has the following properties:

- `title`: Title of the book
- `author`: Author of the book
- `available`: Boolean indicating if the book is available

Include the following methods:

- `borrow()`: Marks the book as not available
- `return\_book()`: Marks the book as available

Then, define a subclass `ReferenceBook` that extends `LibraryBook`. A reference book can never be borrowed. Modify the `borrow()` method to print 'This book cannot be borrowed.'

5. (10 marks) Polymorphism: Shopping Cart System

Define abstract classes `Product` and `Electronics`. Then, implement the following concrete classes:

- **Book**: Costs 100 Baht, with a 10% discount on all purchases.
- **TV** (inherits from `Electronics`): Costs 8,000 Baht with a 7% VAT.
- **Laptop** (inherits from `Electronics`): Costs 20,000 Baht with a 7% VAT.

Write a program to calculate the total cost of the following items:

- 2 Books
- 1 TV
- 1 Laptop

Hint: Use polymorphism to implement a `calculate_cost()` method in each class.